Mitochondrial matrix proteins carry no signature of a 50 °C environment

A probe-independent test of the "hot mitochondria" hypothesis using public thermal-proteome data, and a proposed one-experiment resolution

Paras Chopra · paras@paraschopra.com · Bangalore, India · October 2026 · Executive summary (4 pages). Full technical report, code and intermediate results: github.com/paraschopra/hot-mitochondria.

In one paragraph. Chrétien et al. (2018) and Terzioglu et al. (2023) reported that the mitochondrial matrix sits at ~50 °C, 10–15 °C above the cytosol, as a steady state. Across 13 species, median proteome melting temperature (Tm) rises ~0.55–0.81 °C per °C of growth temperature. A matrix that is permanently 10–15 °C hotter should therefore have proteins melting ~5–12 °C above their cytosolic neighbours. Reanalysing the Meltome Atlas (62 datasets, 4 labs, 6 eukaryotic species, 126,176 Tm measurements), we find matrix − cytosol ΔTm = **−0.17 °C** in lysates (intrinsic stability; 95% CI −1.00 to +0.58) and **+1.26 °C** in intact cells (+0.72 to +1.77). Same-family mitochondrial/cytosolic paralog pairs show no difference. A protein-language-model thermostability predictor shows no heat-adaptation signature in any of five species. A persistently hot matrix is compatible with these data only if an unobserved, matrix-specific, non-intrinsic stabilisation worth ~5–12 °C exists. We propose a single TPP experiment (respiration on vs off in intact cells) that would decide between the two pictures.

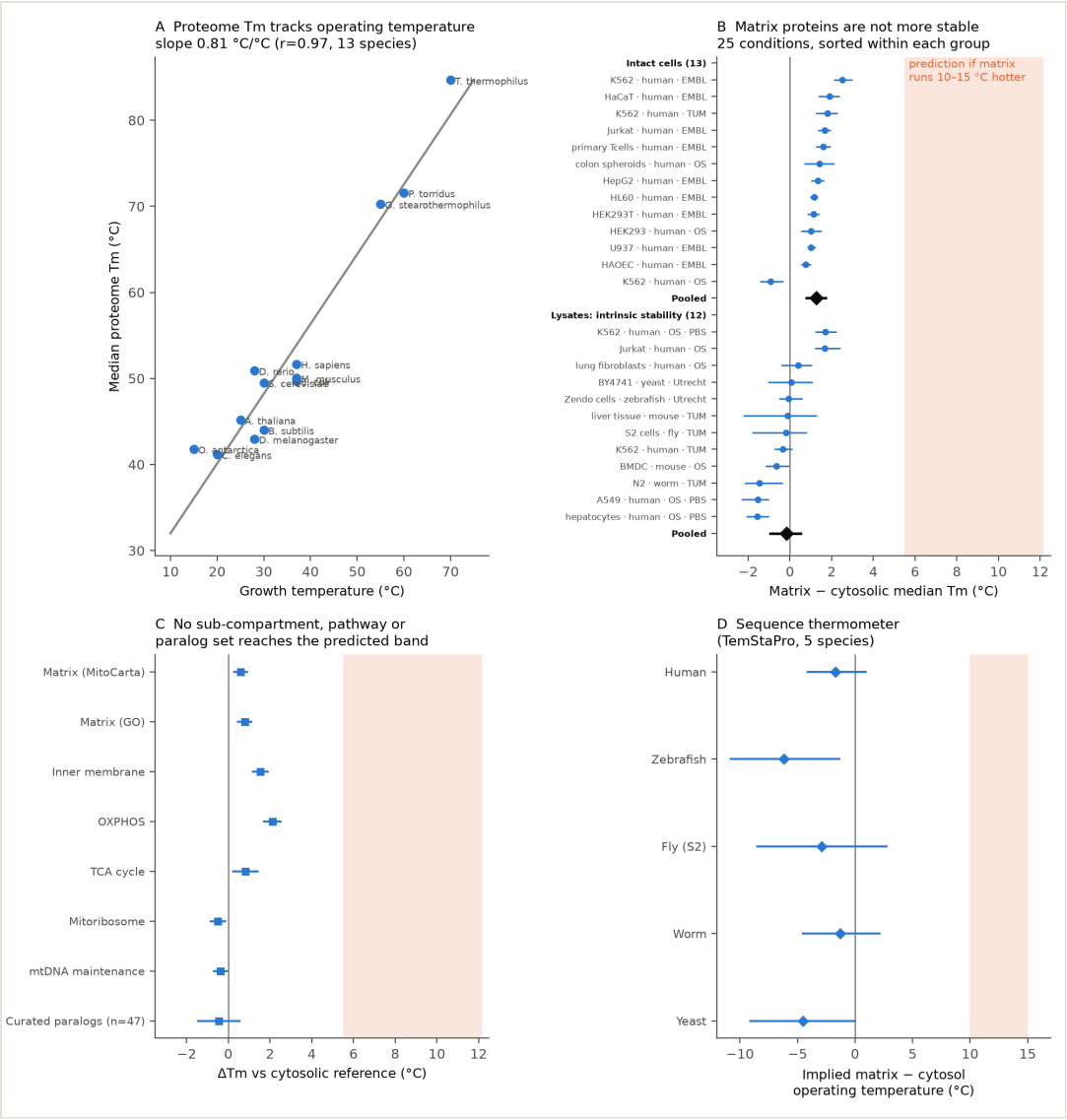
Why this question

The hot-mitochondria claim remains openly contested in 2025–2026: critiques recommending rejection (Treberg & Mailloux 2026, *Acta Physiol*; Tkachenko et al. 2026 preprint), defences (Fahimi, Lynch & Matta 2026, *BioEssays*; Rustin et al. 2025, *TIBS*), and neutral reviews calling it "a matter of intense debate" (Beignon et al. 2026). Every existing measurement uses a fluorescent or diamond thermometer with known confounders (membrane potential, binding, pH, extrapolated calibration). Physics objects to the size of the gradient (the "10^5 gap"), though a 2026 study reports that intracellular heat dissipates non-diffusively over seconds, reopening that argument (Takarada et al. 2026, *Nat Commun*). What has been missing is a test that relies on neither the probes nor heat-transfer theory: whether the matrix proteins themselves show the adaptation a hot environment would select for.

Key results

Test	Matrix – cytosol	95% CI	Prediction if matrix is +10–15 °C
Lysates: intrinsic stability (12 conditions)	**−0.17 °C**	−1.00, +0.58	+5.5 to +12
Intact cells (13 conditions)	**+1.26 °C**	+0.72, +1.77	+5.5 to +12
Non-mammals: worm, fly S2, zebrafish, yeast	−0.41 °C	−1.33, +0.34	+5.5 to +12
47 curated mito/cyto paralog pairs (human)	−0.45 °C	−1.51, +0.60	+5.5 to +12
Sequence thermometer (TemStaPro/ProtT5), human	−1.7 °C*	−4.2, +1.0	+10 to +15*
Inner-membrane proteins: lysates / intact cells	+0.18 / +2.2 °C	−1.2, +1.7 / +1.4, +2.8	+5.5 to +12
Respiratory chain (OXPHOS) subunits: lysates / intact cells	+1.1 / +2.8 °C	−0.6, +2.5 / +1.9, +3.9	+5.5 to +12
Mitoribosome / mtDNA maintenance	−0.5 / −0.4 °C	both < 0	+5.5 to +12

*Implied operating-temperature difference. Positive control passes: human proteins read hotter than worm, fly, fish and yeast orthologs. 0 of 5 species show matrix > cytosol. For the lysate result to be consistent with +10 °C, the Tm–temperature coupling would need to be <0.06 (observed 0.55–0.81; species bootstrap $P \leq 0.006$).



A Proteome Tm vs growth temperature, 13 species. **B** Matrix – cytosol ΔTm in 25 conditions (cell · species · lab), grouped by intact cells / lysates; diamonds = pooled. **C** Alternative definitions, sub-compartments, pathways, paralogs. **D** Sequence thermometer, five species. Orange bands = prediction for a 10–15 °C hotter matrix.

What is new relative to prior work

The closest prior work is Jacobs et al. 2024 (*FEBS J*, by the hypothesis's proponents). In a supplementary analysis of intact K562/Jurkat TPP data they noted mitochondrial proteins are ~2 °C more stable than the proteome median and read this as *support*. We agree with the observation and add:

1. **The expected effect size.** Calibrated across 13 species, a 10–15 °C hotter compartment predicts +5–12 °C, not +2. The matrix median Tm (~51 °C) would sit at the claimed operating temperature, i.e. zero safety margin, unlike any organism measured.
2. **Intrinsic vs environmental split.** In lysates the gap is ~0. The small in-cell gap is environmental, not sequence adaptation.
3. **Within-family control** (47 curated paralog pairs, localisation-verified).
4. **An assay-independent line**: sequence-encoded thermostability, with a positive control.
5. **Six species, four labs**, including *Drosophila* S2 cells, in which hot mitochondria were also claimed.

Searches of PubMed, Europe PMC full text (~318 citing papers of Chrétien 2018), bioRxiv, arXiv and Google Scholar found no controlled proteome-wide, paralog or sequence-based test. Thermal profiling with respiratory inhibitors has been done only in lysates (Cheng et al. 2025) or isolated mitochondria (Rivera-Mejías et al. 2024), neither of which can detect a hot matrix.

Exact claim, and what it is not

> **Claim.** The matrix proteome shows no intrinsic thermal adaptation relative to the cytosol (in measured stability, in same-family pairs, or in sequence-encoded signatures) across six species. A persistently 10–15 °C hotter matrix is therefore compatible with the data only if an unobserved matrix-specific, non-intrinsic stabilisation worth ~5–12 °C exists, or if probe-reported "temperature" is not a steady bulk temperature.

> **Not claimed.** We did not measure mitochondrial temperature, and we don't claim an upper bound of 1–2 °C. The test concerns a *steady* excess (sustained over minutes to evolutionary time). It does not address transient, picosecond-scale heat spikes at the inner membrane (Fahimi, Lynch & Matta 2026), which would leave no imprint on protein stability; even proteins embedded in the inner membrane show no adaptation in lysates (table above). The inference assumes that the cross-species Tm–temperature coupling transfers, at least roughly, to compartments within a cell. This is the central stated assumption.

How the analysis was checked

- Replicated across 4 labs, 25 conditions, 6 species; no condition above +2.5 °C. Two independent matrix definitions (MitoCarta 3.0, GO).
- Uncertainty from a two-level bootstrap (gene clusters shared across all datasets and species, plus conditions). Calibration uncertainty propagated by species bootstrap. Censoring-aware rank test treating non-melters as most stable: P(matrix > cytosol) = 0.53.
- Non-human labels via strict one-to-one orthologs; paralog pairs hand-curated with localisation checks.
- Adversarial review by an independent AI model (OpenAI Codex). It found real errors (orthology, replicate pooling, CI construction, invalid pairs) and conceptual over-claims. All were fixed, and the headline was narrowed accordingly.
- The analysis was carried out with AI assistance (Anthropic's Claude). All code, inputs and intermediate outputs are available for inspection.

Limitations

- TPP Tm measures 3-min aggregation, not equilibrium unfolding. It is, however, the same measure used for the calibration.
- Intact-cell Tm is recorded in dish-temperature coordinates. If mitochondria were hot during heating, intact-cell values would be biased; hence our main inference rests on lysates, paralogs and sequence.
- Fly data are lysate-only; no bird or warm-bodied-fish meltome yet exists for a within-animal positive control.

Key references

Chrétien D et al. (2018) PLoS Biol 16:e2003992 · Terzioglu M et al. (2023) eLife 12:RP89232 · Jarzab A et al. (2020) Nat Methods 17:495 · Jacobs HT et al. (2024) FEBS J · Rath S et al. (2021) MitoCarta3.0, NAR 49:D1541 · Pudžiuvelytė I et al. (2024) TemStaPro, Bioinformatics · Treberg JR & Mailloux RJ (2026) Acta Physiol · Fahimi P et al. (2026) BioEssays · Baffou G et al. (2014) Nat Methods 11:899 · Lane N (2018) PLoS Biol 16:e2005113 · Savitski MM et al. (2014) Science 346:1255784 · Takarada M et al. (2026) Nat Commun 17:4215 · Rivera-Mejías P et al. (2024) bioRxiv 10.1101/2024.02.27.582308 · Tkachenko A et al. (2026) Research Square 10.21203/rs.3.rs-8744427/v1.

Proposed decisive experiment

Design. Thermal proteome profiling in intact cells in two states: **respiring** vs **respiration blocked**, the latter in two forms: antimycin A alone (membrane potential maintained by ATP-synthase reversal on glycolytic ATP; to be confirmed by TMRM) and antimycin A + oligomycin (potential collapses). Optional third arm: uncoupled (FCCP), which by the hypothesis's logic should be hotter still (UCP1 uncoupling increased the MTY "temperature" in Chrétien 2018). Cells: HEK293 or K562 (human), plus *Drosophila* S2. Standard 10-temperature TPP (37–67 °C), TMTpro or label-free DIA, 2 biological replicates.

Logic. TPP records the *dish* temperature at which each protein aggregates. If respiring mitochondria are 10 °C hotter than the dish, matrix proteins reach their true Tm at a dish temperature ~10 °C lower. Blocking respiration removes the excess, so **matrix proteins (only) should appear ~5–10 °C more stable**. Cytosolic proteins, at dish temperature in both states, serve as the internal control. If mitochondria are not hot, matrix and cytosolic proteins shift by the same small amount (<1–2 °C). The two outcomes differ by an order of magnitude more than TPP precision (~0.5 °C).

Outcome	Matrix ΔTm (off – on)	Cytosol ΔTm	Interpretation
Hot mitochondria	+5 to +10 °C	≈ 0	Steady hot matrix confirmed, plus a large unknown protective mechanism to explain the lysate data
Not hot	≈ cytosol (<1–2 °C)	≈ 0	Steady +10 °C matrix ruled out; probe signals have non-thermal components
FCCP arm (if hot)	more negative than "on"	≈ 0	Direction check: the effect should reverse sign

Essential controls. Respirometry (Seahorse/Oroboros) confirming respiration is blocked and that ATP and viability are maintained over the ~1 h treatment. Membrane-potential readout (TMRM). Ligand-sensitive proteins whose Tm shifts with ATP/NADH flagged and analysed separately, since inhibitors change metabolite levels; the test statistic is the matrix-specific shift *relative to* cytosolic proteins with similar ligand sensitivity. Mitochondrial dysfunction can shift mitochondrial protein Tm non-thermally (Groh et al. 2023), so we also require the heat-only signature: a near-uniform shift across all matrix protein classes, plus a parallel lysate control. Loss of membrane potential itself destabilises matrix proteins (antimycin on isolated mitochondria: 299 of 893 proteins destabilised, mostly matrix; Rivera-Mejías et al. 2024). That effect runs *opposite* to the hot-matrix prediction, so the two "off" arms are compared to separate heat from membrane potential.

Scale. ~1–2 weeks of bench work for a lab running TPP. Roughly 60 MS runs (3 states × 2 replicates × 10 temperatures) per cell line. Indicative cost for a label-free pilot in India ≈ ₹3 lakh (~US$3.5k); a full TMT design ≈ ₹10–13 lakh (~US$12–15k).

What we are looking for

- **A TPP lab** to run (or co-design) the experiment; with co-authorship. We are open to discussing support for consumables and MS time.
- **Mitochondrial physiologists**, on either side of the debate, to sanity-check design and controls before data are collected. Pre-registering predictions jointly would make the result persuasive whichever way it falls.
- **Critical reading** of the reanalysis itself. We'd value being told where it is wrong.

Further material

1. Full technical report (methods, all numbers, verification ladder, review log): ~10 pages.
2. Code and intermediate results: Python scripts reproducing every number from public data (Meltome Atlas, MitoCarta 3.0, UniProt, Alliance orthology, TemStaPro).
3. A draft experimental protocol for the respiration on/off TPP with predictions written down in advance.